

ARNAV MALHOTRA

Student Researcher & Engineer

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EXPERIENCE

Wharton Global High School Investment Competition

Global

Lead Quantitative Researcher

Winner (1/6300 Globally)

- Placed in the top 1 out of 6300+ teams globally. Engineered a multi-asset quantitative strategy utilizing an ensemble of Black-Litterman models, multi-objective optimization, and an ML-based crisis prediction framework for robust risk mitigation.

Regeneron

Mentorship Student

Mentorship Student - Regeneron Genetics Center

- Analyzed the construction of SAM alignment files, which store DNA sequences and corresponding error probability information, to identify storage inefficiencies. Working on implementing tests on real alignment data to determine the viability and reward of fixing such inefficiencies.

The Stuyvesant Spectator

New York, NY

Web Lead

- Maintain the digital infrastructure for an award-winning high school publication. Lead the web department in optimizing content delivery, preserving site reliability, and ensuring continuous uptime.

AI Clinical Decision Support Tool

New York, NY

Software Developer

- Working with a team under the mentorship of Dr. Priyank Jain to create a generative AI tool to aid nurses in rural parts of the world. Three pipeline stages produce a triage decision, prescription, and risk-assessed management plan that a remote doctor reviews asynchronously.

RESEARCH

Gordian Engine

Pareto Analysis Machine

- A framework execution engine that systematically isolates and maps theoretical problems and bottlenecks in any field. It outputs an objective ranking of the most valuable, important, and costly problems in the field, dictating the exact constraints that must be expanded.

fsynth: Synthetic Financial Universes

Python Package, Preprint

- Traditional machine learning models trained on financial market data are known to be highly susceptible to overfitting the past market. Historical data is limited, and models trained only on data from the past will only be able to perform well in financial conditions mimicking the past. To combat this data scarcity, many use mathematical generators of unlimited synthetic data. However, traditional synthetic data generators often rely on Geometric Brownian Motion (GBM), which fails to account for empirical phenomena such as volatility clustering, leverage effects, and "black swan" jump events.
- This paper introduces fsynth, a multi-factor synthetic engine that integrates a Heston-based stochastic volatility model with Merton jump diffusion, governed by a hidden Markov regime-switching process. By using macro-regime states alongside company specific "genes," the engine generates plausible universes containing both price action and fundamental accounting data. We demonstrate that while the engine can be calibrated to replicate conditions of historical benchmarks, its primary utility lies in its parameterizable nature, allowing for the simulation of arbitrary, "out-of-sample" stress scenarios.

Quantum Binomial Options Hedging Framework

Preprint

- The Quantum Binomial Options Pricing Model is a novel approach to pricing derivatives. It serves as a quantum analogue to the classical binomial options pricing model, using concepts from quantum mechanics like the Pauli matrices and Bloch sphere to represent the derivatives market. Through usage of a density matrix and mixed states, the model can potentially price derivatives more accurately. However, at this time there exist no clear formulae and methods for hedging derivatives through this model.

- This paper numerically tests the performance of the hedging framework corresponding to the model as compared to the classical model using a year of historical S&P 500 data. The simulation reveals that the Quantum Binomial Options Pricing model is, at least in its current state, not viable for usage in actually hedging derivatives. This is due to numerical instabilities, particularly in the Quantum Theta and Quantum Vega, the former of which ends up negative, betraying fundamental financial postulates, and the latter of which is blown out of scale. We hypothesize that these nonsensical values for Quantum Theta and Quantum Vega stem from the single period nature of the model we tested, as well as the isotropic assumption of the operator A. We conclude that the single period Quantum Binomial Options Pricing Model is currently unsuitable for practical applications of hedging and that our findings reveal the necessity of rigorously testing new financial models against benchmark models like the Classical Binomial Options Pricing Model.

PROJECTS & OPEN SOURCE

Contrepa | *Smart LLM Codebase Context Generator* | contrepa.com

- Using Contrepa is very simple. Paste in a link to a GitHub repository, and load it in. You will visually see the entire tree, and you have two options: fit to a specified token count, or manually select items. With either one, different files will either be copied, summarized by an LLM, or omitted entirely. Summaries of files are created entirely in the browser; the user can choose to input an API key or have the LLM run on their own GPU.

EVE Online Market Arbitrage | *Commodity Analytics*

- Developed an intraregional arbitrage calculator and predictive commodity modeling system for the Jita market, analyzing synthetic macroeconomic behavior in massive multiplayer environments.

EDUCATION

Stuyvesant High School

New York, NY

Class of 2028

- Courses include differential equations, linear algebra, AP Physics C, AP Computer science, AP Calculus BC, AP Environmental Science, AP Statistics, and AP Human Geography.